

Polynomial Linear Regression

Importing Libraries

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Import datasets

```
In [3]: dataset = pd.read_csv("01_dataset/Position_Salaries.csv")
X = dataset.iloc[:,1:-1].values
y = dataset.iloc[:, -1].values
```

```
In [4]: print(dataset)
```

	Position	Level	Salary
0	Business Analyst	1	45000
1	Junior Consultant	2	50000
2	Senior Consultant	3	60000
3	Manager	4	80000
4	Country Manager	5	110000
5	Region Manager	6	150000
6	Partner	7	200000
7	Senior Partner	8	300000
8	C-level	9	500000
9	CEO	10	1000000

```
In [5]: print(X)
```

```
[[ 1]
 [ 2]
 [ 3]
 [ 4]
 [ 5]
 [ 6]
 [ 7]
 [ 8]
 [ 9]
[10]]
```

```
In [6]: print(y)
```

```
[ 45000  50000  60000  80000 110000 150000 200000 300000 500000
1000000]
```

Training the Linear Regression model on the whole dataset

```
In [7]: from sklearn.linear_model import LinearRegression
lin_reg = LinearRegression()
lin_reg.fit(X, y)
```

```
Out[7]: ▼ LinearRegression ⓘ ⓘ
LinearRegression()
```

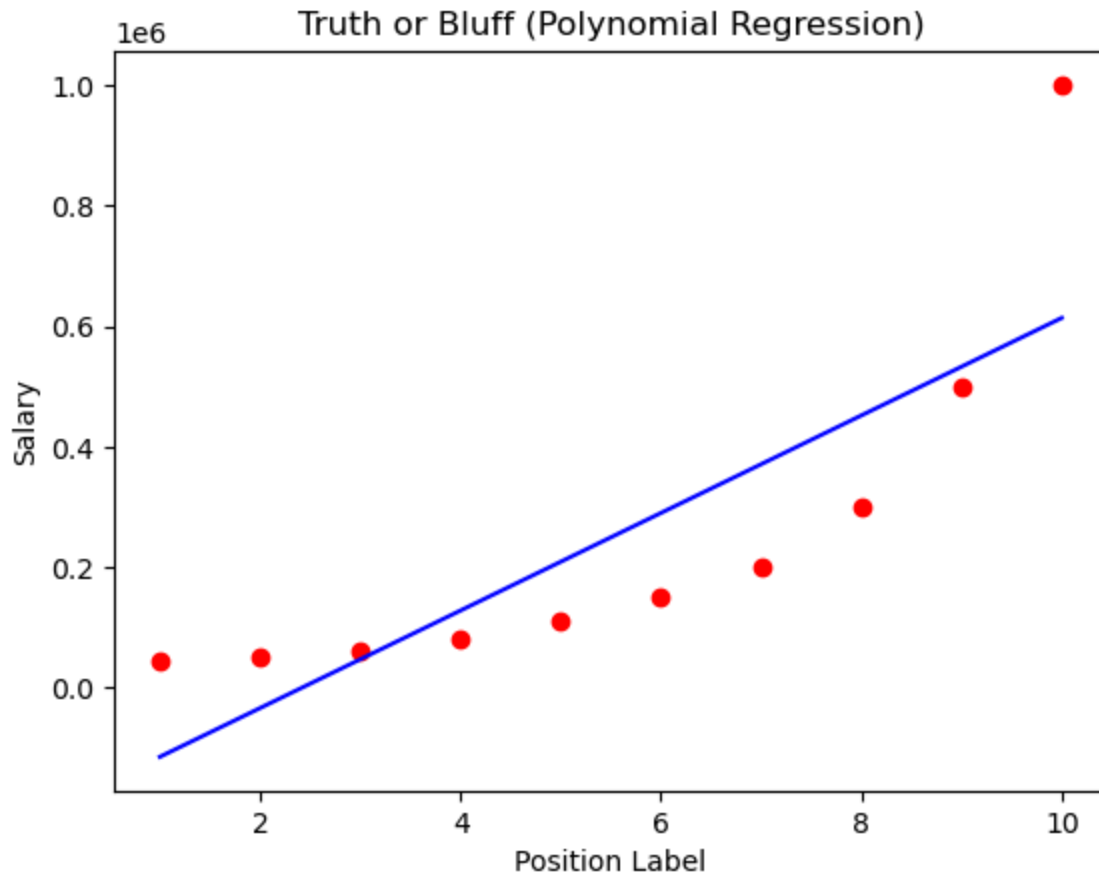
Training the Polynomial Regression model on the whole dataset

```
In [8]: from sklearn.preprocessing import PolynomialFeatures
poly_reg = PolynomialFeatures(degree = 4)
X_poly = poly_reg.fit_transform(X)
lin_reg_2 = LinearRegression()
lin_reg_2.fit(X_poly, y)
```

```
Out[8]: ▼ LinearRegression ⓘ ⓘ
LinearRegression()
```

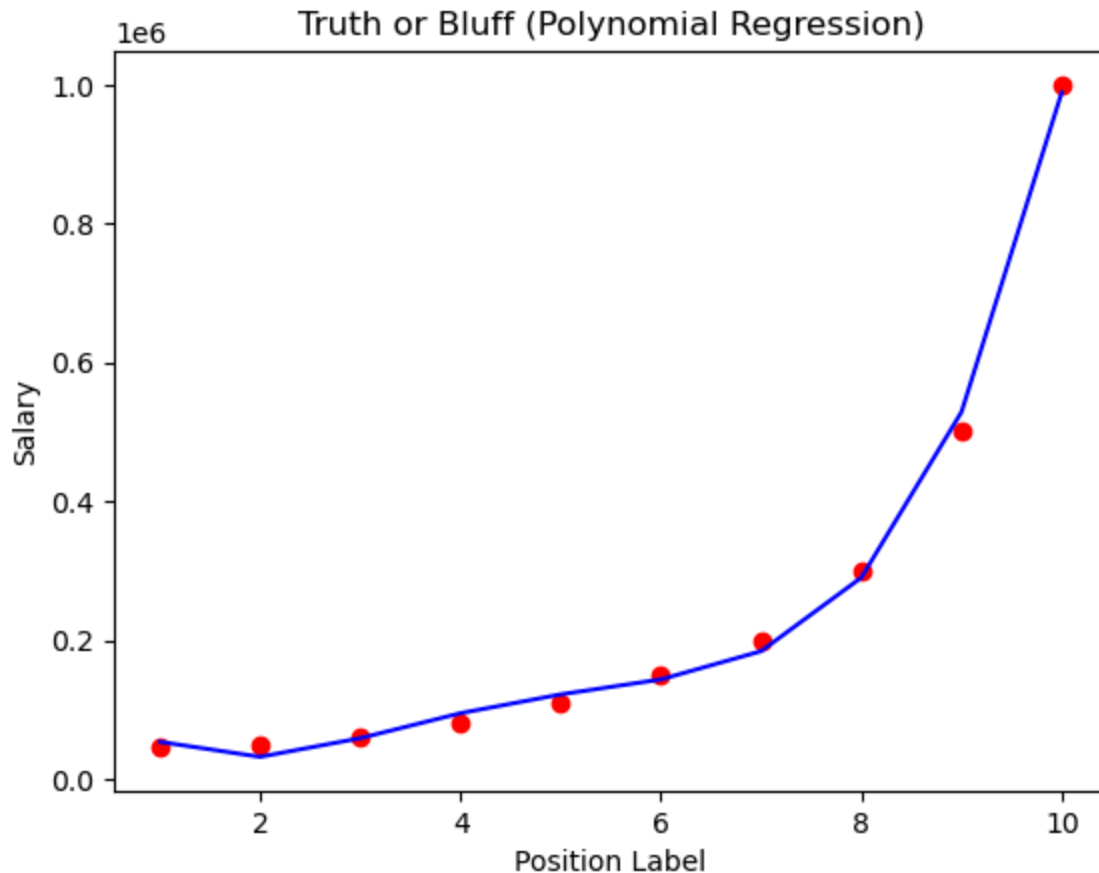
Visualising the Linear regression results

```
In [13]: plt.scatter(X, y, color = 'red')
plt.plot(X, lin_reg.predict(X), color = 'blue')
plt.title("Truth or Bluff (Linear Regression)")
plt.xlabel("Position Label")
plt.ylabel("Salary")
plt.show()
```



Visualising the Polynomial Regression results (for higher resolution and smoother curve)

```
In [16]: plt.scatter(X, y, color = 'red')
plt.plot(X, lin_reg_2.predict(X_poly), color = 'blue')
plt.title("Truth or Bluff (Polynomial Regression)")
plt.xlabel("Position Label")
plt.ylabel("Salary")
plt.show()
```

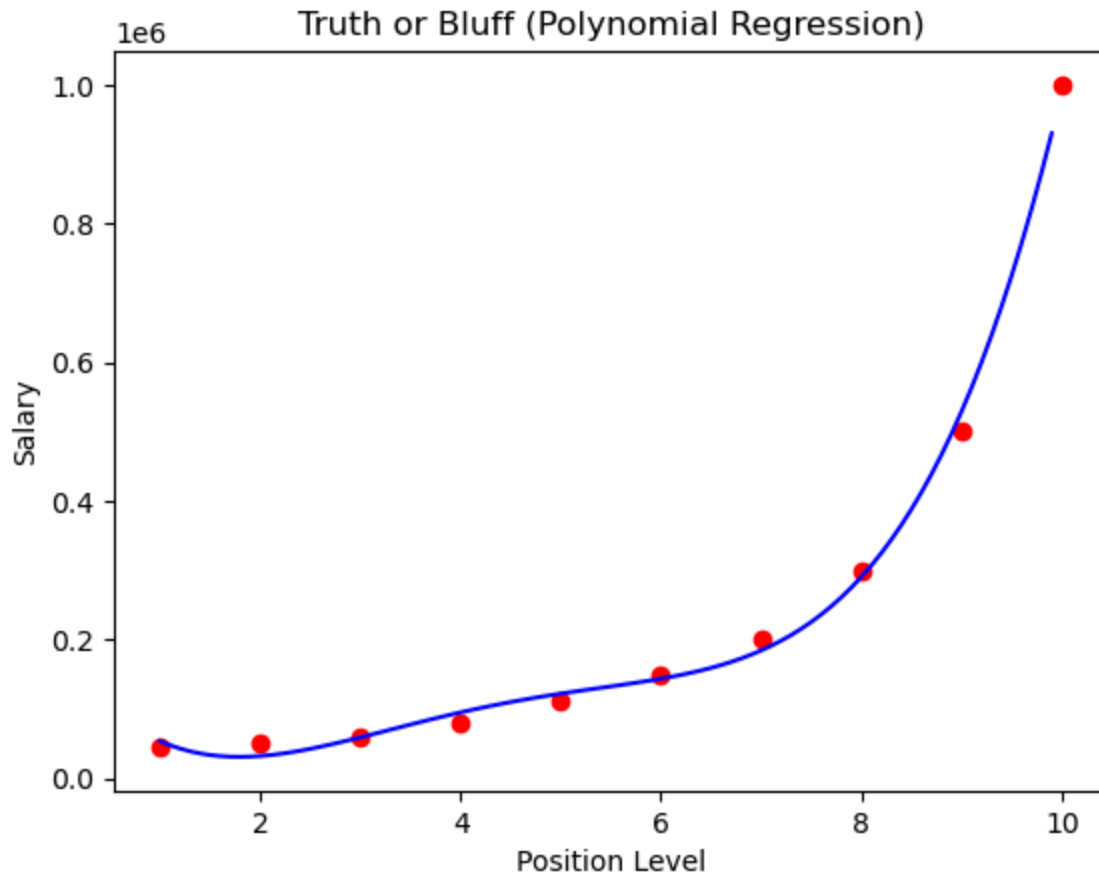


Visuaising the Polynomial Regression results (for higher resolution and smoother curve)

```
In [17]: X_grid = np.arange(min(X), max(X), 0.1)
X_grid = X_grid.reshape((len(X_grid), 1))
plt.scatter(X, y, color = 'red')
plt.plot(X_grid, lin_reg_2.predict(poly_reg.fit_transform(X_grid)), color =
plt.title('Truth or Bluff (Polynomial Regression)')
plt.xlabel('Position Level')
plt.ylabel('Salary')
plt.show()
```

/tmp/ipykernel_9263/3390505073.py:1: DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is deprecated, and will error in future. Ensure you extract a single element from your array before performing this operation. (Deprecated NumPy 1.25.)

```
X_grid = np.arange(min(X), max(X), 0.1)
```



Predicting a new result with Linear Regression

```
In [18]: lin_reg.predict([[6.5]])
```

```
Out[18]: array([330378.78787879])
```

Predicting a new result with Polynomial Regression

```
In [20]: lin_reg_2.predict(poly_reg.fit_transform([[6.5]]))
```

```
Out[20]: array([158862.45265155])
```